

A 185 dB FoM_s Direct-Integration Third-Order Passive Noise-Shaping SAR ADC for Biomedical Applications

Kangkang Sun , Kailin Lin , Weijie Ge , Feng Yan , Bingjun Xiong , Jian Guan, Jingjing Liu*

School of Electronics and Communication Engineering, Sun Yat-sen University, Shenzhen Campus, Shenzhen 518107, China.

Email: liujj77@mail.sysu.edu.cn

Abstract—This paper proposes a high-energy-efficiency, direct-integration third-order passive noise-shaping (NS) successive approximation register analog-to-digital converter (SAR ADC) for biomedical applications. By directly sampling the residue voltage on the integrator and connecting the integration capacitors to the comparator through a charge-pump cascade, the proposed NS SAR ADC significantly reduces quantization-noise loss during the sampling/integration process, relaxes the gain requirement of the differential input pairs, lowers comparator power consumption, and enhances noise-shaping performance. The proposed NS SAR ADC is designed based on a standard 180nm CMOS technology. Under a 1.2V power supply, the ADC achieves 87.36 dB signal-to-noise-and-distortion ratio (SNDR) over a 55.6 kHz bandwidth at 1.78 MS/s while consuming only 9.67 μ W. The corresponding Schreier figure-of-merit (FoM) is 184.96 dB and the Walden FoM is 4.6 fJ/conversion-step, offering an energy-efficient solution for ultra-low-power biomedical systems.

Keywords—SAR ADC, Third-Order Passive Noise-Shaping, Charge-pump cascade, Multi-input comparator, Biomedical Applications

I. INTRODUCTION

Driven by the synergistic advancement of biomaterial science, nanofabrication technology, and ultra-low-power integrated circuit design, wearable and implantable medical electronic systems with excellent biocompatibility and ergonomic adaptability are playing an increasingly crucial role in continuous health monitoring and precision disease therapy[1][2][3]. Within the physiological signal acquisition chain, ADC serves as the quantization interface between multi-channel biosensor arrays and the digital backend. Its precision and energy efficiency fundamentally determine the overall system reliability and operational lifespan[4][5][6]. Therefore, achieving high-precision, ultra-low-power ADCs is a critical design bottleneck for extending the battery life of implantable medical devices, particularly in applications requiring continuous, long-term physiological signal acquisition.

NS SAR ADCs combine the noise shaping technique of delta-sigma (Δ - Σ) ADCs with the low-power advantage of SAR ADCs, making them increasingly attractive for low-power, high-precision physiological signal acquisition systems[7][8]. Passive NS SAR ADCs not only eliminate the high power consumption associated with active structures but also reduce

design complexity, thereby demonstrating excellent energy efficiency. However, in switch-capacitor-based passive NS schemes, charge sharing during the integration process leads to signal attenuation, which ultimately degrades the noise transfer function (NTF) and compromises quantization accuracy. Therefore, to compensate for signal loss during the integration process, feedforward NS circuits often employ multi-input comparators that provide one or more high-gain input paths[9][10]. In practical designs, the gain differential pairs, which have gains significantly greater than 1, exhibit substantially larger transistor dimensions and higher power consumption than the unity-gain input pairs. This leads to increased overall power dissipation, a larger layout area, and higher input-referred noise.

To reduce the gain of the multi-input comparator and optimize the noise shaping performance of the integrator, this paper proposes a direct-integration third-order noise-shaping SAR ADC. This architecture directly samples the quantization residue voltage from the CDAC through the integration capacitors, which not only eliminates the need for noise sampling capacitors but also reduces noise loss and kT/C noise throughout the process. Subsequently, the first-, second-, and third-order integration capacitors are connected to the comparator input in a charge-pumped cascaded configuration, significantly reducing the comparator input gain and further lowering the overall power consumption.

The remainder of this paper is organized as follows: Section II introduces the direct-integration third-order passive noise-shaping architecture; Section III discusses the noise-shaping performance and stability; Section IV presents the simulated performance metrics of the NS SAR ADC. And at last, Section V concludes the paper.

II. DIRECT-INTEGRATION THIRD-ORDER PASSIVE NOISE-SHAPING ARCHITECTURE

To mitigate the high-gain requirement of multi-input comparators and optimize integrator-based noise-shaping performance, this paper proposes a direct-integration third-order noise-shaping SAR ADC. As illustrated in Fig.1, the architecture consists of third-order passive noise integrator, a reduced-gain multi-input comparator, SAR logic, CDAC, and bootstrapped switches. This architecture simplifies the two processes of noise sampling and noise integration into a direct noise integration process. As a result, it not only saves timing

This work was supported by Shenzhen Science and Technology Program under Grant JCYJ20250604175601003 and ZDCY20250901104259009.

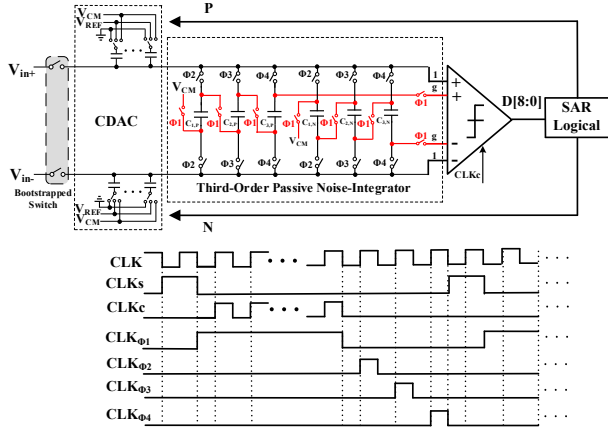


Fig. 1. The proposed direct-integration third-order passive noise-shaping SAR ADC architecture and its timing scheme.

and area resources but also reduces noise loss and kT/C noise. Simultaneously, the third-order integration capacitors are connected to the comparator input in a charge-pump cascade configuration, thereby further reducing the required input-stage gain and power consumption.

When $CLK_{\phi 1}$ is high, the basic quantizer of the 9-bit SAR ADC completes the quantization; immediately afterward, to extract the residue V_{RES} , the DAC capacitor array is flipped conditionally according to $D[0]$. Then, by sequentially controlling the switch states of capacitors $C_{1,P/N}$, $C_{2,P/N}$, and $C_{3,P/N}$, the first-order, second-order, and third-order noise integration of the residue voltage are completed.

When $CLK_{\phi 2}$ is high, the top plates of $C_{1,P}$ and $C_{1,N}$ are connected to the top plate of the CDAC P-side array, while their bottom plates are tied to the top plate of the CDAC N-side array. At this instant, the residue voltage stored on the CDAC is shared with $C_{1,P}$ and $C_{1,N}$, effecting a first-order passive sample-and-integrate operation. Denoting the integrated voltage on $C_{1,P}$ and $C_{1,N}$ as V_{INT1} and assuming $C_{1,P}=C_{1,N}$, charge conservation applied to $C_{1,P}$ yields:

$$V_{INT1}(n-1) \cdot 4C_{1,P} + V_{RES}(n-1) \cdot C_{DAC} = V_{INT1}(n) \cdot (4C_{1,P} + C_{DAC}) \quad (1)$$

where $V_{INT1}(n-1)$ denotes the integration voltage stored on $C_{1,P/N}$ at the end of the previous quantization cycle, and $V_{RES}(n-1)$ is the residue voltage generated during that cycle. Defining the capacitive ratio $a=C_{1,P/N}/C_{DAC}$ and transforming the charge-conservation equation into the z -domain yield the first-order integrated voltage on $C_{1,P}$ and $C_{1,N}$:

$$V_{INT1}(z) = \frac{V_{RES}(z)z^{-1}}{4a+1-4az^{-1}} \quad (2)$$

For $a = 0.5$, equation (2) simplifies to

$$V_{INT1}(z) = \frac{V_{RES}(z)z^{-1}}{3-2z^{-1}} \quad (3)$$

Similarly, it can be derived that:

$$V_{INT2}(z) = \frac{V_{INT1}(z)}{3-2z^{-1}} \quad (4)$$

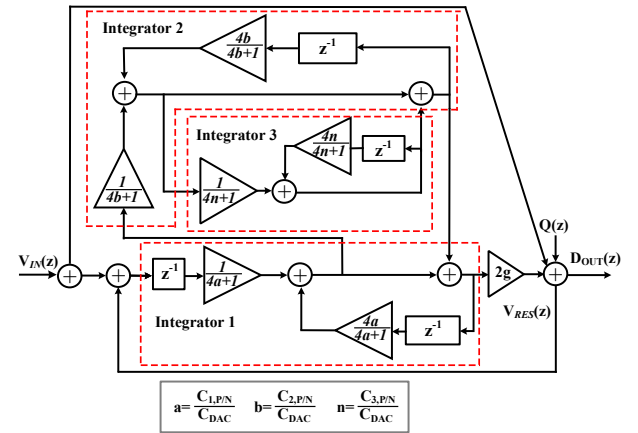


Fig. 2. Signal flowchart analysis of the direct-integration third-order passive NS SAR ADC.

$$V_{INT3}(z) = \frac{V_{INT2}(z)}{2-z^{-1}} \quad (5)$$

By scaling the aspect ratio (W/L) of the V_{INT} input differential pair to g -times that of the CDAC top-plate voltage input pair, a voltage gain of g is conferred upon V_{INT} , exactly offsetting the attenuation experienced during integration. Constructing the signal-flow graph from the established noise-shaping signal paths yields the complete system diagram shown in Fig. 2. Denoting the integrated signal, quantization noise, and residue voltage as indicated, the governing relations are:

$$V_{RES}(n) = -2g(V_{INT1}(n) + V_{INT2}(n) + V_{INT3}(n)) + Q(z) \quad (6)$$

Substituting the expressions for $V_{INT1}(z)$, $V_{INT2}(z)$, and $V_{INT3}(z)$ yields:

$$V_{RES}(z) = -2g \left(\frac{V_{RES}(z)z^{-1}}{3-2z^{-1}} + \frac{V_{RES}(z)z^{-1}}{(3-2z^{-1})^2} + \frac{V_{RES}(z)z^{-1}}{(3-2z^{-1})^2(2-z^{-1})} \right) + Q(z) \quad (7)$$

Rearranging, we obtain:

$$NTF = \frac{V_{RES}(z)}{Q(z)} = \frac{18-33z^{-1}+20z^{-2}-4z^{-3}}{18+(18g-33)z^{-1}+(20-16g)z^{-2}+(4g-4)z^{-3}} \quad (8)$$

III. NOISE-SHAPING PERFORMANCE ANALYSIS

Equation (8) shows that the NTF is a function of the comparator input gain g . Therefore, the value of g determines the poles of the NTF, which in turn affects the stability of the NTF system and the noise shaping performance. Consequently, a systematic trade-off analysis of the value of g is required to achieve the optimal balance between energy consumption and performance.

A. NTF Stability Analysis

According to Expression (8), the numerator of the NTF is a third-order polynomial in terms of z . By solving the equation $18-33z^{-1}+20z^{-2}-4z^{-3}=0$, its zeros are found to be: $1/2$, $2/3$, and $2/3$. Theoretically, zeros do not affect system stability. For the denominator of the NTF, it is a third-order polynomial containing the parameter g . To determine the poles of the NTF, we solve the equation $18+(18g-33)z^{-1}+(20-16g)z^{-2}+(4g-4)z^{-3}=0$, yielding three poles:

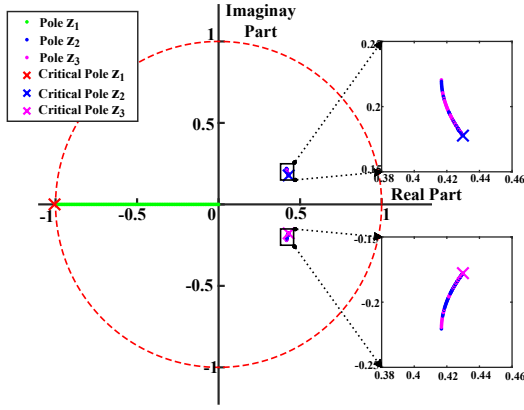


Fig. 3. Migration trajectories of poles z_1 , z_2 , and z_3 as g increases.

$$z_1 = \frac{\frac{1}{18}\sqrt[3]{-216g^3 + 324g^2 + 12\sqrt{3}\sqrt{72g^4 - 116g^3 + 50g^2 + g - 162g - 1}}}{\frac{-324g^2 + 324g - 9}{162\sqrt[3]{-216g^3 + 324g^2 + 12\sqrt{3}\sqrt{72g^4 - 116g^3 + 50g^2 + g - 162g - 1}} + \frac{1}{18}(11 - 6g)}} \quad (9)$$

$$z_2 = \frac{\frac{1}{36}(1 - i\sqrt{3})\sqrt[3]{-216g^3 + 324g^2 + 12\sqrt{3}\sqrt{72g^4 - 116g^3 + 50g^2 + g - 162g - 1}}}{\frac{(1 + i\sqrt{3})(-324g^2 + 324g - 9)}{324\sqrt[3]{-216g^3 + 324g^2 + 12\sqrt{3}\sqrt{72g^4 - 116g^3 + 50g^2 + g - 162g - 1}} + \frac{1}{18}(11 - 6g)}} \quad (10)$$

$$z_3 = \frac{\frac{1}{36}(1 + i\sqrt{3})\sqrt[3]{-216g^3 + 324g^2 + 12\sqrt{3}\sqrt{72g^4 - 116g^3 + 50g^2 + g - 162g - 1}}}{\frac{(1 - i\sqrt{3})(-324g^2 + 324g - 9)}{324\sqrt[3]{-216g^3 + 324g^2 + 12\sqrt{3}\sqrt{72g^4 - 116g^3 + 50g^2 + g - 162g - 1}} + \frac{1}{18}(11 - 6g)}} \quad (11)$$

Since the poles z_1 , z_2 , and z_3 are all functions of the gain g , their positions in the z -plane are determined solely by the selected gain. To ensure the stability of the NTF system, all three poles must lie within the unit circle. By modeling the expressions of z_1 , z_2 , and z_3 , it can be observed that when $g > 1$, the trajectories of the three poles within the unit circle as g increases are shown in Fig. 3. Among them, pole z_1 exhibits the highest sensitivity to g . As g increases, z_1 monotonically decreases along the real axis and reaches the critical pole value of -1 at approximately $g \approx 1.9798$. Poles z_2 and z_3 are both located in the right half of the complex plane, with their real parts gradually increasing and the absolute values of their imaginary parts progressively decreasing as g increases. Overall, poles z_2 and z_3 exhibit minimal variation with changes in g , whereas the trajectory of pole z_1 along the real axis determines the allowable range of g .

B. Performance Analysis of the NTF

The proposed third-order passive integrator delivers an out-of-band noise rise of 60 dB/decade and, as the oversampling ratio (OSR) increases, exhibits strong low-frequency suppression, ultimately achieving about -20 dB of in-band noise

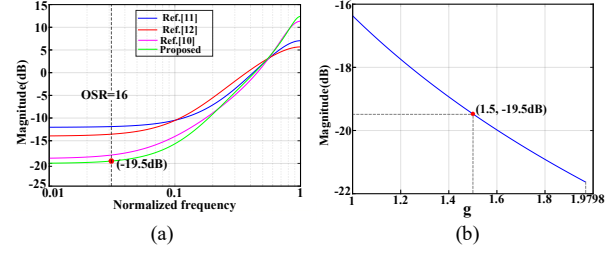


Fig. 4 NTF magnitude–frequency characteristics: (a) Magnitude response of the NTF at $g = 1.5$. (b) NTF magnitude versus g at OSR = 16.

attenuation. As shown in Fig. 4(a), under identical conditions, prior passive noise-shaping implementations report inferior performance: Ref. [11] ($g_{c2} = g_{a2} = 2$) reaches -11.9 dB at OSR = 16; Ref. [12] (unity gain) attains only -13.6 dB at OSR = 16; Ref. [10] (gain budget $g_1 = 18$, $g_2 = 1.6$) achieves -18.1 dB at OSR = 16. With the gain budget set to 1.5, the NTF derived in this work achieves a demonstrably superior power-performance trade-off, simultaneously minimizing power dissipation while maximizing in-band noise attenuation.

To quantitatively assess the dependence of NTF magnitude on gain g , the proposed NTF is modeled and simulated at an oversampling ratio of OSR = 16. As illustrated in Fig. 4 (b), the NTF magnitude decreases monotonically with increasing g , signifying continuously enhanced low-frequency suppression. Based on a comprehensive trade-off among comparator power dissipation, noise-shaping efficacy, and overall loop stability, an integration gain of $g = 1.5$ is selected as the optimum operating point.

C. Design of the comparator circuit

The comparator employed in this work is illustrated in Fig. 5. To reduce input-referred noise and enhance comparison speed, it consists of a pre-amplifier followed by a dynamic latch. To avoid mismatches among multiple tail-current sources typically associated with multi-input pairs [13], a single shared tail transistor is adopted. The relative gain g between the two input pairs is precisely set by their aspect ratio (W/L), enabling accurate current-domain summation and comparison in a single step.

Similar to the Strong-Arm dynamic comparator, the dual-input dynamic comparator never creates a DC path from supply to ground; hence, only dynamic power is consumed during operation, which is the key to its low-power capability.

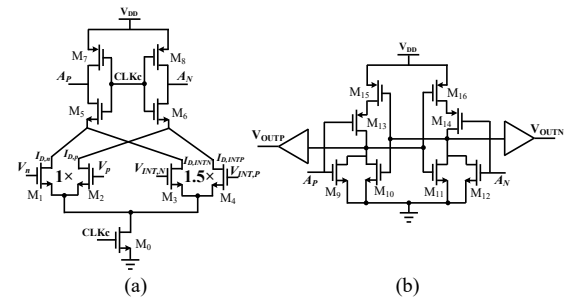


Fig. 5. Dual-input-pair dynamic comparator: (a) pre-amplifier, (b) dynamic latch.

IV. SIMULATION RESULTS

The proposed ADC is designed based on a 180 nm CMOS process with a supply voltage of 1.2 V. Schematic simulation results show that: when the sampling frequency is 1.78 MHz, the input sinusoidal signal has an amplitude of -0.087 dBFS and a frequency of 5.01 kHz, with a sampling number of 8192, the output frequency spectrum characteristics of the designed ADC are presented in Fig. 6. Operating at $OSR = 16$, the ADC achieves a signal bandwidth of 55.6 kHz, with a computed SNDR of 87.36 dB, an effective number of bits (ENOB) of 14.22 bit, and an spurious-free dynamic range (SFDR) of 97.24 dB. The variation of SNDR and SFDR with input amplitude is summarized in Fig. 7. During normal operation the ADC consumes 9.67 μ W. As illustrated in Fig. 8, the power is distributed as follows: digital logic 31.48 %, clock generation 4.64 %, CDAC 52.21 %, comparator 6.72 %, and others 4.95 %. Under these conditions the Schreier FoM reaches 184.96 dB, while the Walden FoM is 4.6 fJ/conversion-step.

Table I summarizes the performance of the proposed NS SAR ADC and compares it with state-of-the-art SAR and NS

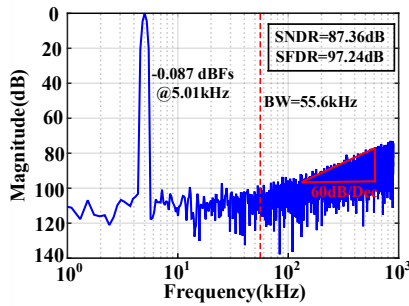


Fig. 6. Simulated spectrum of the proposed ADC.

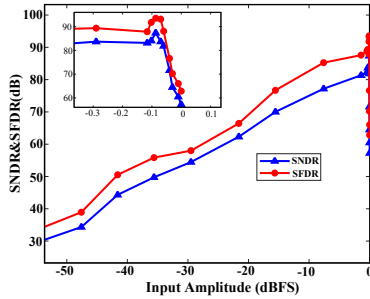


Fig. 7. SNDR and SFDR versus input amplitude.

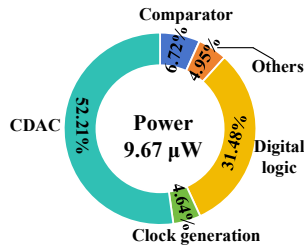


Fig. 8. Power breakdown.

TABLE I. PERFORMANCE SUMMARY AND COMPARISON

	This work*	ISCAS** 25 [16]	ISCAS** 24 [17]	ISSCC* 23 [14]	TCASI* 24[15]
Process(nm)	180	40	55	180	180
Supply(V)	1.2	1.1	1	1	1
Architecture	NS-SAR			SAR	
Order	3	2	2	2	-
F_s (MHz)	1.78	50	0.4	0.128	8
OSR	16	8	32	32	-
BW(kHz)	55.6	3125	6.25	1	4000
Resolution	9	10	12	5	16
SNDR(dB)	87.36	81.8	91.05	99.1	85.7
SFDR(dB)	97.24	-	97.9	106.8	97.7
Power(μ W)	9.67	190.3	7.426	9.29	9500
FoM _W (fJ/Conv.-step)	4.6	2980	20.4	63	76 000
FoM _S (dB)	184.96	184	180.3	179.4	175.4

*Simulation results

FoM_W= Power/(2^{ENOB}*2*BW)

FoM_S= SNDR+10lg(BW/Power)

SAR ADCs. Benefiting from the minimal noise loss during integration provided by the direct-integration third-order passive noise-shaping architecture and the reduced comparator power achieved by the charge-pump series connection of integrating capacitors, the proposed ADC consumes lower power and achieves significantly better FoM_S than other 180 nm implementations [14][15]. Compared with ADCs designed in more advanced process [16][17], this work not only delivers higher FoM_S but also exhibits lower FoM_W. These results demonstrate that the proposed NS SAR ADC is well suited for biomedical electronic systems with stringent low-power and high-precision requirements.

V. CONCLUSION

This paper presents a low-power, direct-integration third-order passive noise-shaping SAR ADC for biomedical applications. The architecture employs integrating capacitors to perform first-, second-, and third-order noise integration directly on the quantization residue remaining on the CDAC, and subsequently connects these capacitors to a multi-input comparator through a charge-pump cascade. Providing a gain of only 1.5 $\times$, the multi-input comparator realizes approximately 20 dB of noise attenuation. The ADC in 180 nm CMOS achieves 87.36 dB SNDR at 1.78 MHz sampling with $OSR = 16$, while consuming 9.67 μ W from a 1.2 V supply. The corresponding Schreier FoM is 184.96 dB and the Walden FoM is 4.6 fJ/conversion-step.

REFERENCES

- [1] A. Paul et al., "A Versatile In-Ear Biosensing System and Body-Area Network for Unobtrusive Continuous Health Monitoring," IEEE Trans. Biomed. Circuits Syst., vol. 17, no. 3, pp. 483-494, June 2023, doi:10.1109/TBCAS.2023.3272649.
- [2] K. Sun et al., "A 0.6-V 9.38-Bit 6.9-kS/s Capacitor-Splitting Bypass Window SAR ADC for Wearable 12-Lead ECG Acquisition Systems," in

- IEEE Transactions on Very Large Scale Integration (VLSI) Systems, vol. 33, no. 7, pp. 1838-1847, July 2025, doi: 10.1109/TVLSI.2025.3559669.
- [3] W. Mao, Y. Li, C.-H. Heng, and Y. Lian, "A low power 12-bit 1-kS/s SAR ADC for biomedical signal processing," IEEE Trans. Circuits Syst. I, Reg. Papers, vol. 66, no. 2, pp. 477-488, Feb. 2019, doi: 10.1109/TCSI.2018.2859837.
 - [4] H. Sun, K. Sun, Y. Wang, W. Mo, F. Yan and J. Liu, "A Second-Order Noise-Shaping SAR ADC for Biomedical Sensor Applications," 2024 IEEE Biomedical Circuits and Systems Conference (BioCAS), Xi'an, China, 2024, pp. 1-5, doi: 10.1109/BioCAS61083.2024.10798387.
 - [5] L. Fang et al., "A 107.6 dB-DR Three-Step Incremental ADC for Motion-Tolerate Biopotential Signals Recording," IEEE Trans. Biomed. Circuits Syst., vol. 18, no. 1, pp. 111-122, Feb. 2024, doi:10.1109/TBCAS.2023.3312976.
 - [6] K. Sun et al., "A 9.68 nW 57.51dB SNDR SAR ADC with Dual Bypass Windows Based on Non-binary Split Capacitors for Biomedical Applications," in IEEE Transactions on Biomedical Circuits and Systems, doi: 10.1109/TBCAS.2025.3557241.
 - [7] Z. Chen, M. Miyahara and A. Matsuzawa, "A 9.35-ENOB, 14.8 fJ/conv.-step fully-passive noise-shaping SAR ADC," 2015 Symposium on VLSI Circuits (VLSI Circuits), Kyoto, Japan, 2015, pp. C64-C65, doi: 10.1109/VLSIC.2015.7231329.
 - [8] L. Jie et al., "An Overview of Noise-Shaping SAR ADC: From Fundamentals to the Frontier," in IEEE Open Journal of the Solid-State Circuits Society, vol. 1, pp. 149-161, 2021, doi: 10.1109/OJSSCS.2021.3119910.
 - [9] H. Zhuang et al., "A Second-Order Noise-Shaping SAR ADC With Passive Integrator and Tri-Level Voting," in IEEE Journal of Solid-State Circuits, vol. 54, no. 6, pp. 1636-1647, June 2019, doi:10.1109/JSSC.2019.2900150.
 - [10] Y. Zhou et al., "A Second-Order Noise Shaping SAR ADC With Parallel Multiresidual Integrator," in IEEE Transactions on Very Large Scale Integration (VLSI) Systems, vol. 32, no. 11, pp. 2135-2138, Nov. 2024, doi: 10.1109/TVLSI.2024.3447740.
 - [11] H. Zhang, Y. Zhu, C.-H. Chan and R. P. Martins, "An Inherent Gain Error Tolerance Noise-Shaping SAR-Assisted Pipeline ADC With Code-Counter-Based Offset Calibration," in IEEE Journal of Solid-State Circuits, vol. 57, no. 5, pp. 1480-1491, May 2022, doi: 10.1109/JSSC.2021.3111912.
 - [12] H. Zhang, Y. Zhu, R. P. Martins and C.-H. Chan, "A Second-Order NS Pipelined SAR ADC With Quantization-Prediction-Unrolled Gain Error Shaping and Fully Passive Integrator," in IEEE Journal of Solid-State Circuits, vol. 58, no. 12, pp. 3565-3575, Dec. 2023, doi: 10.1109/JSSC.2023.3307189.
 - [13] Masaya Miyahara, Yusuke Asada, Daehwa Paik and Akira Matsuzawa, "A low-noise self-calibrating dynamic comparator for high-speed ADCs," 2008 IEEE Asian Solid-State Circuits Conference, Fukuoka, Japan, 2008, pp. 269-272, doi: 10.1109/ASSCC.2008.4708780.
 - [14] G. Kim et al., "32.4 A 1V-Supply 1.85VPP -Input-Range 1kHz-BW 181.9dB-FOMDR179.4dB-FOMSND 2nd-Order Noise-Shaping SAR-ADC with Enhanced Input Impedance in 0.18 μ m CMOS," 2023 IEEE International Solid-State Circuits Conference (ISSCC), San Francisco, CA, USA, 2023, pp. 484-486, doi: 10.1109/ISSCC42615.2023.10067844.
 - [15] S. Huang et al., "An 8-MS/s 16-bit SAR ADC With Symmetric Complementary Switching and Split Passive Reference Segmentation in 180-nm Process," in IEEE Transactions on Circuits and Systems I: Regular Papers, vol. 71, no. 10, pp. 4486-4498, Oct. 2024, doi: 10.1109/TCSI.2024.3430378.
 - [16] Y. Ai, Y. Guo, Z. Lou, H. Zhang, J. Jin and J. Zhou, "An EF-CIFF Noise-Shaping SAR ADC with A Joint Dynamic Amplifier, Comparator and Lossless Passive Summer Structure," 2025 IEEE International Symposium on Circuits and Systems (ISCAS), London, United Kingdom, 2025, pp. 1-5, doi: 10.1109/ISCAS56072.2025.11044022.
 - [17] Y. Lu et al., "A 91 dB SNDR Calibration-Free Fully-Passive Noise-Shaping SAR ADC with Mismatch Error Shaping," 2024 IEEE International Symposium on Circuits and Systems (ISCAS), Singapore, Singapore, 2024, pp. 1-5, doi: 10.1109/ISCAS58744.2024.10557948.